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29 June 1979
SUPERSEDING
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MILITARY SPECIFICATION

TRAILER, FLAT-BED: 10 TON, 4 WHEEL M345

This specification is approved for use by US Army Tank-Automotive Materiel Readiness Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a tactical flat-bed, 10 ton, 4 wheel type trailer designated as the M345.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

VV-B-680	- Brake Fluid, Automotive
VV-L-800	- Lubricating Oil, General Purpose, Preservative, (Water-displacing, Low-Temperature).
ZZ-I-550	- Inner Tube, Pneumatic Tire.

MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-L-3150	- Lubricating Oil, Preservative, Medium.
MIL-H-3992	- Hose and Hose Assembly, Rubber: Air and Vacuum Brake, Automotive.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Materiel Readiness Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 2330

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MIL-G-10924 - Grease, Automotive and Artillery.
MIL-T-12459 - Tire, Pneumatic: for Military Ground Vehicles.
MIL-H-13719 - Hose Assembly, Rubber; Hydraulic Brake.
MIL-H-13910 - Hydraulic Fluid, Polar Type, Automotive Brake, All-weather.
MIL-G-23827 - Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-V-62038 - Vehicles, Wheeled: Preparation for Shipment and Storage of.

STANDARDS

FEDERAL

FED-STD-595 - Color.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-193 - Finishing Procedures, Tactical Vehicles (Tracked and Wheeled).
MIL-STD-642 - Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-810 - Environmental Test Methods.
MIL-STD-1179 - Lamps, Reflectors and Associated Signaling Equipment for Military Vehicles.

DRAWINGS

ARMY

8358914 - Trailer, Flat-Bed: 10 Ton, 4 Wheel, M345.

SUPPLEMENTARY QUALITY ASSURANCE PROVISIONS

MASTER LIST

8358914 - Trailer, Flat-Bed: 10 Ton, 4 Wheel, M345.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 First article (preproduction) vehicle. When specified (see 6.2), a preproduction model of the complete vehicle to be furnished under this specification shall be produced prior to the manufacture or fabrication of vehicles in quantity. The vehicle, when completed, shall be submitted to the Government for examination and tests to determine conformance to the requirements of this specification. Vehicle submitted by the contractor shall be fully representative of vehicles proposed to be supplied from production facilities and tooling under the supply contract.

3.2 Initial production vehicles. The initial production vehicles selected shall be fully representative of vehicles proposed to be furnished under contract, with all current modifications included. Submitted vehicles shall be examined and tested to determine conformance to the requirements of this specification. No modifications shall be made to subsequent vehicles produced without prior Government approval.

3.2.1 Reliability. The design goal of reliability for the M345 shall be 6,000 Mean Miles Between Failure (MMBF), to be demonstrated in accordance with the operation profile during test. For calculation of MMBF, a failure shall be defined as a condition which:

a. Prevents operation, or
b. Reduces performance below essential levels, or
c. Indicates to the crew that further operation would be unsafe, or
d. Indicates to the crew that further operation might result in extensive damage to the equipment.

Any of the foregoing conditions which cannot be corrected by the crew in 30 minutes, using the tools and parts normally carried on the M345 or the tow vehicle, constitutes a failure. Maintenance and human induced failure are excluded.

3.2.1.1 Operational profile. Test mileage shall be apportioned as follows:

a. 40 percent hard-surfaced roads
b. 25 percent gravel and dirt roads
c. 23 percent level cross-country
d. 2 percent belgian block
e. 10 percent hilly cross-country

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3.2.2 Maintainability. Total maintenance, excluding driver/crew checks and services, shall not exceed 15 manhours during 6,000 miles of operation (see 3.2.1.1). This equates to a maintenance ratio of .075 at 30 operational miles, equivalent to one operational hour. The scheduled maintenance interval shall be 3 months or 3,000 miles whichever comes first.

3.3 Materials. Materials shall be as specified herein and in referenced specifications and drawings. Materials shall be free from all defects and imperfections that might affect serviceability and appearance of the finished product (see 6.6).

3.3.1 Qualified products. The contractor shall be responsible for using parts and assemblies from Qualified Products Lists (QPL's) whenever available. Contractor's inspection records shall specifically list all QPL items by number and date of the QPL, name of supplier and part or drawing number(s). When parts and assemblies are approved as qualified products, but not yet listed on the QPL, the contractor shall list the products by number and date of the approved document and date of the approved document and name of supplier(s).

3.3.2 Ozone resistance. When rubber components are required to be ozone resistant, the degree of resistance shall be as specified in applicable specifications or drawings.

3.4 Construction. Trailer components, subassemblies and assemblies shall be fabricated and assembled into a complete trailer in accordance with drawing 8358914. All parts, assemblies, and subassemblies requiring identification shall be identified in accordance with MIL-STD-130.

3.4.1 Seals. When fording or operating in mud, sand or snow, seals shall prevent entrance of foreign matter into the bearings, which are exposed to contamination during these operations. All bearings seals shall restrict the leaking of lubricants from bearings.

3.4.2 Electrical.

3.4.2.1 Electrical circuits. Electrical circuits shall maintain continuity from end to end without evidence of internal or external shorts during all trailer operating conditions.

3.4.2.2 Lights. All lights as specified on applicable drawings shall operate throughout all trailer operating conditions and shall comply with the requirements of MIL-STD-1179.

3.4.3 Controls. All electrical, mechanical, and hydraulic controls shall operate without malfunction, throughout all ranges of operation, under all trailer operating conditions.

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3.4.4 Adjustment mechanisms. All adjustment mechanisms shall function properly and maintain adjustment settings during all vehicle operating conditions.

3.4.5 Welding. Welding shall meet the quality requirements specified on the applicable drawing, or Supplementary Quality Assurance Provisions (SQAP's).

3.4.5.1 Welding repairs. Welding repairs of any kind shall be made only when, and to the extent, specifically authorized by the procuring activity (see 6.2).

3.4.6 Tires and tubes.

3.4.6.1 Tires. Tires shall be class CC, style "a", conforming to MIL-T-12459.

3.4.6.2 Inner tubes. Inner tubes shall conform to group 2 of ZZ-I-550 and be of proper size for the tire specified.

3.5 Performance. Trailer shall meet performance requirements of this specification when equipped as specified herein, serviced with standard products specified in table I, loaded with rated payload and coupled to the tactical tow vehicle specified in table II. Trailer, serviced and equipped for existing climatic conditions, shall operate as specified without special equipment.

TABLE I. Service products specifications.

Product use	Ambient air temperatures	
	-65°F to 0°F	-10° to + 115°F
Oil:		
For hydraulic brakes	MIL-H-13910	VV-B-680
For general purpose lubrication	VV-L-800	MIL-L-3150
Grease:		
For sealed bearings	MIL-G-23827	MIL-G-23827
For general bearings including wheel bearings	MIL-G-10924	MIL-G-10924

3.5.1 Environmental. The trailer shall operate under extreme conditions of weather and altitude in ambient air temperatures ranging from minus 65°F to plus 115°F. The complete trailer when in storage shall withstand a temperature of minus 80°F, without deterioration that may cause failure of any component.

3.5.2 Payload, towing speeds, and trailing ability.

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3.5.2.1 Highway (paved) and gravel roads, and speeds. The trailer shall be loaded with highway rated payload, or with a simulated load of equal weight, uniformly distributed over load area. Trailer shall be towed over relatively level, smooth, improved and prepared hard-surfaced roads and gravel roads at average speeds specified in table II, without damage to the trailer tow vehicle. The trailer shall track the tow vehicle in a straight line on a level, smooth, paved surface without deviating more than 3 inches to either side of the path of the tow vehicle.

TABLE II. Loads and speeds.

Trailer	Tactical tow vehicle	Payload (pounds)		Speed (miles per hour)		
		Highway	Cross country	Highway	Gravel roads	Cross country
M345	M809 series or similar tow vehicle	26,000	20,000	50	20	20 Max

3.5.2.2 Cross-country payload and speeds. The trailer, loaded with cross-country rated payload uniformly distributed over the load area, shall be capable of being towed over unimproved roads, trails, open fields, rolling hills and cross-country terrain at average speeds specified in table II, without damage to trailer or tow vehicle. The payload shall be distributed so that the proper center of gravity is established.

3.5.2.3 Belgian block without payload. The 6,000 mile road test shall include 50 miles of gravel and 30 miles of Belgian block without payload at start and completion of highway and cross-country operations, for a total of 160 miles.

3.5.3 Grade and slope operation. The trailer shall operate without slipping or upsetting when ascending or descending longitudinal grades up to 40 percent and side slopes up to 20 percent.

3.5.4 Brake holding ability.

3.5.4.1 Service brakes. The combined service brakes of the trailer and tow vehicle shall, under all condtions of loading, bring the combination to a positive stop within a distance of 30 feet from a speed of 20 miles per hour on dry, smooth and level road, free of loose material. Brake hose and brake hose assemblies shall meet the requirements of MIL-H-3992 and MIL-H-13719, as applicable.

3.5.4.2 Service braking system. The trailer air reservoir shall be charged with air from 100 to 105 pounds per square inch (psi). After disconnection of the brake air supply line, service brakes shall:

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a. Be applied to all wheels of the trailer.
b. Not display hydraulic fluid leakage.
c. Maintain air pressure with a drop of no less than 1/2 psi per minute and no more than 5 psi per minute.

3.5.4.3 Automatic actuation. Trailer shall be equipped with an automatic actuating device to apply the trailer brakes upon breakaway from the tow vehicle. When loaded with rated highway payload, the automatic actuating device shall maintain application of the brakes and hold trailer stationary on a 20 percent grade for not less than 15 minutes.

3.5.5 Fording ability. When towed through fresh or salt water to a depth great enough to entirely submerge the chassis for periods up to 15 minutes duration, vehicle or its components shall sustain no damage. This shall be accomplished without the use of special kits and with a minimum of preparation or servicing before and after fording operations.

3.5.6 Landing legs. Each landing leg shall raise, support and hold its portion of the load and shall withstand the strain imposed, without damage, when coupling or uncoupling the tow vehicle. Each landing leg shall extend and retract; and when retracted, shall clear the ground in cross-country operation.

3.5.7 Turning ability. When coupled to the tow vehicle and operating in a minimum turning circle, the trailer shall follow without cramping, damage or interference between the trailer and the tow vehicle.

3.5.8 Wheel and tire balancing. Each assembly of wheel, tube and tire shall be in balance to preclude wheel tramp or drumming at road speeds up to 50 mph.

3.6 Painting, marking and identification plates.

3.6.1 Treatment and painting. Treatment and painting shall be in accordance with MIL-STD-193. Color shall conform to FED-STD-595.

3.6.2 Marking. Trailer markings shall be in accordance wtih applicable drawing and MIL-STD-642. Component parts marking shall conform to MIL-STD-130.

3.6.3 Identification plates. Identification plates shall meet the requirements of MIL-P-514.

3.7 Safety. All exposed parts which are energized electrically shall be located, insulated, fully enclosed or guarded to prevent hazards to operating personnel and equipment functioning. All moving parts which are of such nature, or so located, as to be a hazard to operating or maintenance personnel shall be enclosed or guarded. Protective devices shall not impair operating functions.

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3.8 Servicing and adjusting. Prior to acceptance by the Government, contractor shall adjust and service each trailer for immediate operational use, including at least the following: Adjust braking system, examine electrical system, inflate all tires and completely lubricate the trailer and all running gear with grades of lubricants recommended for the ambient temperature at the delivery point (see table I).

3.9 Workmanship. Workmanship shall be of such quality as to assure that the trailer and its components are free of defects that compromise, limit or reduce the capability of the trailer in the performance of its intended use. In addition to general appearance, the defects listed in tables V and VI shall be considered to be cause for failure.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, or specifically disapproved by the Government the contractor is responsible for the performance of all inspection requirements specified herein and may use his own or any other facilities suitable for the performance of the inspection requirements. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Contractor's quality assurance system. The contractor shall provide and maintain an effective inspection and quality assurance system, acceptable to the Government, covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor shall not be restricted to the inspection listed, provided that an equivalent control is included in the approved quality assurance procedure. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that might affect the degree of assurance required by this specification, or other applicable documents referenced therein.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of surveilliance of the operations to determine that practices, methods, and procedures of the written inspection plan are being properly applied, and Government product inspection to measure quality of product offered for acceptance. Deviation from prescribed or agreed-upon procedures, or instances of poor practices which might have an adverse effect upon the quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies shall be cause for suspension of acceptance until corrective action has been made, or until conformance of product to prescribed criteria has been demonstrated.

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4.1.3 Materials and qualified products. The contractor's inspection records shall be examined to determine contractor conformance to 3.3 through 3.3.2.

4.1.4 Parts and components. Parts, components and assemblies shall be inspected in accordance with drawings listed on drawing 8358914 and Master List of Supplementary Quality Assurance Provisions drawing 8358914.

4.2 Classification of inspections. Classification of inspections shall be as follows:

a. First article (preproduction) inspection (see 4.3).
b. Initial production vehicle inspection (see 4.4).
c. Quality conformance inspection (see 4.5).

4.3 First article (preproduction) inspection. A first article vehicle (see 3.1), when furnished, shall be inspected and tested to determine conformance to the requirements of this specification. Inspection shall be performed by the Government. Failure of the first article vehicle to meet any requirements specified herein shall be cause for refusal of acceptance until corrective measures satisfactory to the Government have been taken and a successful retest has been conducted.

4.4 Initial production vehicle inspection.

4.4.1 In-process inspection. During fabrication of the initial production vehicle (see 3.2), an in-process inspection will be conducted by the Government to evaluate conformance of materials and workmanship to specified requirements. These inspections shall be made at the contractor's or subcontractor's facility prior to application of primer and paint. Processing and welding procedures, quality system and inspection records will be reviewed and evaluated during this inspection.

4.4.2 Completed initial vehicle inspection. The initial production vehicle, on each contract shall be road tested and completely inspected by the contractor for conformance to the requirements of contract and specifications. The applicable road test shall be that specified in 4.7.2.2. Upon completion, the contractor shall submit the vehicle (and make available all inspection records and certifications) to the responsible Government inspection element at contractor's plant for preliminary inspection.

4.4.2.1 Preliminary inspection. The vehicle shall receive a preliminary inspection by the responsible Government inspection element. The preliminary inspection shall be complete, except for road test requirements of 4.7.2.2, which will be conducted jointly during provisional inspection to be accomplished by representatives of US Army Tank-Automotive Materiel Readiness Command, Product Assurance Directorate, as specified in 4.4.2.2. Preliminary inspection shall include in-process inspection as specified in 4.4.1.

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4.4.2.2 Provisional inspection. The first completed production vehicle shall be subjected to provisional inspection at the contractor's plant by representatives of US Army Tank-Automotive Materiel Readiness Command, Product Assurance Directorate. The contractor shall provide any required inspection assistance. At the time of this inspection, the contractor shall make available his inspection plan, inspection records and certifications pertinent to the vehicle and components.

4.4.2.3 Repair of defects. Defects found as a result of the foregoing inspections shall be corrected by the contractor at no cost to the Government. Failure of the contractor to promptly correct defects shall be cause for suspension of vehicle acceptance until corrective action has been accomplished.

4.4.3 Vehicle disposition. After completion of inspection, the initial production vehicle shall remain at the manufacturing facility, as a production sample, and be the last vehicle shipped on the contract. The vehicle may be released sooner at the discretion of the contracting officer. The contractor shall service and maintain the vehicle during this period.

4.4.4 Final approval and acceptance. Final approval and acceptance by the Government of the initial production vehicle shall be withheld until the 6,000 mile test of the second vehicle produced has been completed and a final determination has been made regarding conformity of the vehicle to requirements. This shall include, but not be limited to, workmanship and materials.

4.4.5 Initial production test. To determine conformance to section 3 (inclusive) after completion of the initial production vehicle inspection, a second trailer shall be selected from the first month's production and subjected to examination specified in tables V and VI and all tests specified in table IV. Subsequent to examination and tests, the selected vehicle shall be subjected to a test of 6,000 miles as specified in table III, at a site approved by the Government, which will require a maximum of 90 days. These tests are to be performed by the Government. The contractor shall furnish repair parts, as required, to support the tests at no cost to the Government.

TABLE III. 6,000 mile road test (Government proving ground).

Course	Mileage and speeds
Hard-surfaced roads	2,400 miles at varying speeds up to maximum
Gravel and dirt roads	1,500 miles at varying speeds up to maximum
Level cross-country	1,380 miles at varying speeds up to maximum
Belgian block	120 miles at varying speeds up to maximum
Hilly cross-country	600 miles at speeds applicable to trailer characteristics

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4.4.5.1 Reliability verification. To determine conformance to 3.2.1, a point estimate value equal to or greater than 6,000 miles shall be demonstrated during test. The point estimate will be computed using the cumulative test mileage of all vehicles divided by the cumulative number of chargeable failures of all vehicles.

4.4.5.2 Maintainability verification. To determine conformance to 3.2.2, a Maintenance Ratio (MR) of less than, or equal to, that specified in 3.2.2 must be demonstrated during test. The MR will be calculated using the total cumulative maintenance manhours, with the exception of maintenance induced errors and driver/crew checks and services.

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TABLE IV. Classification and location of inspections and tests.

Title	Require-ments	Init. prod test GPG 1/	Accept test MFG 2/	Control test MFG 2/	Comparison test GPG 1/
Initial production vehicles	3.2	4.4.5			
Reliability verification	3.2.1	4.4.5.1			
Maintainability verification	3.2.2	4.4.5.2			
Seals	3.4.1	4.5.1	4.5.1		4.5.1
Electrical circuits	3.4.2.1	4.5.2	4.5.2		4.5.2
Lights	3.4.2.2	4.5.3	4.5.3		4.5.3
Controls	3.4.3	4.5.4	4.5.4		4.5.4
Adjustment mechanisms	3.4.4	4.5.5	4.5.5		4.5.5
Welding, and welding repairs	3.4.5 and 3.4.5.1	4.5.6	4.5.6		4.5.6
Environmental	3.5.1	4.5.7			4.5.7
Highway (paved), and gravel roads payload and speeds	3.5.2.1	4.5.8	4.5.8	4.5.8	4.5.8
Cross-country payload and speeds	3.5.2.2	4.5.9	4.5.9	4.5.9	4.5.9
Grade and slope operation	3.5.3	4.5.10		4.5.10	4.5.10
Service brakes	3.5.4.1	4.5.11	4.5.11	4.5.11	4.5.11
Service braking system	3.5.4.2	4.5.12	4.5.12	4.5.12	4.5.12
Automatic brake hold	3.5.4.3	4.5.13	4.5.13	4.5.13	4.5.13
Fording ability	3.5.5	4.5.14		4.5.14	
Landing legs	3.5.6	4.5.15	4.5.15	4.5.15	
Turning ability	3.5.7	4.5.16	4.5.16	4.5.16	
Wheel and tire balancing	3.5.8	4.5.17	4.5.17	4.5.17	
Treatment and painting	3.6.1	4.5.18	4.5.18		

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TABLE IV. Classification and location of inspections and tests. - Continued

Title	Requirements	Init. prod test GPG 1/	Accept. test MFG 2/	Control test MFG 2/	Comparison test GPG 1/
Marking	3.6.2	4.5.19			
Identification plates	3.6.3	4.5.20	4.5.20		
Safety	3.7	4.5.21	4.5.21		
Servicing and adjusting	3.8	4.5.22	4.5.22		
Workmanship	3.9	4.5.23			
Preservation, packaging and vehicle processing	Section 5	4.5.24			

1/ Government proving ground.
2/ Manufacturer/contractor facility.

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4.4.5.3 Test failure. Failure of the vehicle to comply with specified requirements, as a result of any defects of a workmanship or materials nature during or as a result of the 6,000 mile test, shall be cause for rejection of the vehicle. Further, the Government may refuse to continue acceptance of production vehicles until evidence has been provided by the contractor that corrective action has been taken to eliminate such defects. Defects found during, or as a result of, the 6,000 mile test shall be prima facie evidence that all vehicles already accepted prior to completion of the test are similarly deficient, unless evidence satisfactory to the Government is furnished by the contractor that they are not similarly defective. Such deficiencies on all vehicles shall be corrected by the contractor at no cost to the Government.

4.5 Conformance verification.

4.5.1 Seals examination. To determine conformance to 3.4.1, after the 5 mile road test, all bearings, as applicable, shall be examined for lubricant leakage.

4.5.2 Electrical circuit examination. To determine conformance to 3.4.2.1, current and voltage output levels and continuity of the electrical system shall be examined.

4.5.3 Lighting system examination. To determine conformance to 3.4.2.2, lights shall be operated and examined for functional requirements.

4.5.4 Controls examination. To determine conformance to 3.4.3, all controls shall be operated and examined for functional requirements.

4.5.5 Adjustment mechanism examination. To determine conformance to 3.4.4, all adjustment mechanisms shall be examined for required adjustment and function.

4.5.6 Welding repairs and burning examination. To determine conformance to 3.4.5 and 3.4.5.1, all welding repairs and burning operations shall be examined to determine that slag has been removed and approved procedures have been followed.

4.5.7 Environmental operation test. To determine conformance to 3.5.1, trailer shall be operated and stored at the temperatures specified. The trailer shall evidence no damage as a result of such operation or storage.

4.5.8 Highway operation test. To determine conformance to 3.5.2.1, the trailer shall be operated as specified on paved and gravel roads and observed for functional requirements.

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4.5.9 Cross-country operation test. To determine conformance to 3.5.2.2, the trailer shall be operated over cross-country terrain as specified without damage to the trailer or the tow vehicle.

4.5.10 Grade and slope operation test. To determine conformance to 3.5.3, the vehicle shall be operated on 40 percent grades headed up and down grade, and on 20 percent side slopes with each side of vehicle up-slope.

4.5.11 Service brakes test. To determine conformance to 3.5.4.1, vehicle combination shall be operated as specified and observed for trailer and tow vehicle stopping distance. Stopping distance shall not be greater than specified.

4.5.12 Service braking system test. To determine conformance to 3.5.4.1 and 3.5.4.2, vehicle shall be operated as specified and system performance observed.

4.5.13 Automatic actuation test. To determine conformance to 3.5.4.3, trailer and tow vehicle shall be placed on the specified grade and the automatic breakaway device acutated. Brakes shall hold trailer stationary for the specified time.

4.5.14 Fording test. To determine conformance to 3.5.5, the trailer shall be towed through water crossings at the depth and for the time period specified.

4.5.15 Landing leg test. To determine conformance to 3.5.6, the trailer shall be placed on firm surface and the landing device and leveling mechanisms operated. The towing vehicle shall be coupled and uncoupled and the leveling mechanism raised and lowered.

4.5.16 Turning and tracking ability test. To determine conformance to 3.5.7, the tow vehicle shall be coupled to the trailer and driven to the tow vehicle's minimum turning circle to the right and to the left, without interference between the tow vehicle and the trailer.

4.5.17 Wheel and tire balance examination. To determine conformance to 3.5.8, wheels and tires shall be observed for unusual noise, wheel-t amp or drumming at road speeds up to 50 mph.

4.5.18 Painting. To determine conformance to 3.6.1, preparation and painting shall be examined during and after application.

4.5.19 Marking. To determine conformance to 3.6.2, vehicle markings shall be examined.

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4.5.20 Identification plates. To determine conformance to 3.6.3, all identification plates shall be examined.

4.5.21 Safety evaluation. To determine conformance to 3.7, the trailer shall be examined to assure that hazards to operating and maintenance personnel have been eliminated or controlled. Equipment which creates hazards shall be considered to be failures.

4.5.22 Servicing and adjusting. To determine conformance to 3.8, servicing and adjustment of specified systems shall be examined.

4.5.23 Workmanship. Trailers shall be examined to determine conformance to 3.9.

4.5.24 Preservation, packaging and vehicle processing inspection. Material and equipment shall be inspected, prior to shipment, to determine conformance to section 5 of this specification.

4.6 Quality conformance inspection.

4.6.1 Inspection provisions.

4.6.1.1 Lot size. An inspection lot shall consist of all vehicles from an identifiable production period (one day's production) submitted at one time for examination and test.

4.6.1.1.1 Sampling for inspection. For the purpose of visual, dimensional, and primary functional inspection, a representative sample shall be selected, from each inspection lot, in accordance with inspection Level II of MIL-STD-105. Before sampling, the contractor shall 100 percent inspect the first 20 trailers to establish a process average (see 6.3), to allow normal sampling in accordance with MIL-STD-105.

4.6.2 Examinations. Visual, dimensional, and primary functional examination shall consist of examination of the vehicle for conformance to applicable drawings and this specification. Examinations shall be performed against the classification of defects and with the Acceptable Quality Levels (AQL) specified in tables V and VI. These examinations shall be performed during all phases of manufacturing and subsequent to road test.

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TABLE V. Major defects AQL - 25 defects/100 units.

Defect number	Characteristics	Major defects	Method of inspection
101	Frame	Structural, welding, riveting, or sheet metal defects	Visual
102	Suspension system components	Damage	Visual
103	Axle	Improper assembly or installation; welding defects	Visual and functional
104	Service brakes; service braking system and automatic brake holding	Malfunction; leaks; component damage; nonconformance to stopping distance; automatic actuation	Visual and funtional
105	Wheels and tires	Wheel or tire damage	Visual
106	Electrical system components and circuits	Malfunction; damage; lighting sequence improper	Visual and functional
107	Electrical wiring, cables, ground straps and lead inlets	Component damage	Visual
108	Leveling jacks	Malfunction: extended, retracted, or stowed	Functional
109	Lunette eye	Casting defects; size improper	Visual
110	Safety chains, intervehicular hose, cables or tubing	Component damage; improper lengths	Visual
111	Tiedown devices	Missing; component damage; welding defects	Visual
112	Drawbar assembly:	Raising or lowering action, improper welding, defects	Visual and functional
113	Landing legs	Locking action improper, raised and lowered; malfunction: extended, retracted or stowed	Functional
114	Tools and equipment	Improper size or type; damage; missing	Visual

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TABLE VI. Minor defects AQL - 150 defects/100 units.

Defect number	Characteristics	Minor defects	Method of inspection
201	Suspension system components	Improper assembly or installation	Visual
202	Brake system components	Improper assembly or installation; clearance inadequate	Visual
203	Wheels and tires	Improper size, type and mounting	Visual
204	Electrical system components	Improper assembly or installation	Visual
205	Electrical wiring cables ground straps and lead inlet	Improper coding, protection, installation, or assembly	Visual
206	Leveling jacks	Improper assembly or installation	Visual
207	Lunette eye	Improper assembly or installation	Visual
208	Landing leg	Improper assembly or installation	Visual
209	Reflectors	Missing, defective	Visual
210	Protective coatings	Application; materials coverage improper	Visual
211	Paint	Improper application or color	Visual
212	Marking, decals, data and instruction plates	Missing; incomplete data; improper size or location	Visual
213	Tube fittings	Defective; missing, improperly installed	Visual
214	Lubrication	Improper application	Visual
215	Hydraulic brake fluid	Improper levels or type	Visual
216	Record forms	Missing; improperly processed	Visual
217	Tools	Missing; improperly processed or secured	Visual

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4.6.2.1 Unclassified defects. All defects that have no effect on function safety, interchangeability or life, but that are considered departures from good workmanship will be noted in writing. Workmanship deficiencies falling within a thirty-day period, will be added to the minor classification with no increase in AQL's (see 6.5).

4.6.2.2 Examination failure. If a vehicle fails to pass any examination or tests specified herein, the Government inspector shall stop acceptance until evidence has been provided by the contractor that corrective action has been taken.

4.7 Classification of tests.

a. Acceptance tests (see 4.7.1).
b. Control tests (see 4.7.2).
c. Comparison tests (see 4.7.3).

4.7.1 Acceptance tests. To determine conformance to section 3 (inclusive), each trailer shall be examined as specified in 4.6.2 and operated for a distance of not less than 5 miles, without payload, by the contractor. Trailers shall be completely assembled and serviced. Acceptance tests shall be those specified in table IV. Performance of trailers shall be demonstrated on relatively smooth, hard-surfaced roads.

4.7.1.1 Test failure. If the vehicle fails to pass any acceptance test specified herein, the Government shall stop acceptance of subsequent vehicles until evidence has been provided by the contractor that corrective actions have been accomplished.

4.7.2 Control tests.

4.7.2.1 Frequency. The Government shall select at random one trailer per month, or one out of each 100 trailers submitted, but not more than two trailers in any given 30 day period, for a 50 mile road test by the contractor.

4.7.2.2 Fifty mile test. Trailer shall be loaded with full or simulated payload, operated for a distance of 50 miles, and subjected to all control tests specified in table IV. These tests shall be performed at the place of manufacture by the contractor. The test course shall be a relatively smooth, hard-surfaced road.

4.7.2.2.1 Test failure. If the vehicle selected fails to pass any control test, the Government shall stop acceptance examination and testing on subsequent vehicles until conditions causing the failure have been remedied. Any defect found during or as a result of the test shall be prima facie evidence that vehicles accepted subsequent to the previous acceptable control test vehicle are similarly defective, unless evidence satisfactory to the contracting officer

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is furnished by the contractor that they are not similarly defective. Such defects on all vehicles shall be corrected by the contractor at no cost to the Government. Another vehicle with corrective actions implemented shall be subjected to the control test.

4.7.3 Comparison tests. The Government may select trailers anytime during the contract production period and subject trailer to all applicable tests listed in table III and IV to reveal deficiencies of workmanship or of a materials nature that may reduce their effective operation in the field and to compare existing quality with previous standards. These tests shall be conducted at Government laboratories or proving grounds designated by the contracting officer. Selection of trailer shall be on a spot check basis. Comparison test trailers shall be loaded as specified in table II, and shall not include any trailer previously tested for conformance to 4.7.2.

4.7.3.1 Test failure. Failure of any vehicle to comply with any of the requirements specified in the contract or any major defects of a workmanship or materials nature, occuring or as a result of a test cycle, may be cause for refusal to continue acceptance of vehicles until objective evidence has been provided by the contractor that corrective action has been taken. Any defects found during, or as a result of the test, shall be prima facie evidence that vehicles accepted subsequent to the previous acceptable comparison test vehicles are similarly defective, unless evidence satisfactory to the contracting officer is furnished by the contractor ~~that they are not similarly~~ defective. Such defects on all vehicles shall be corrected by the contractor at no cost to the Government.

5. PACKAGING

5.1 Preservation, processing. Preservation, processing for shipment and storage shall be in accordance with MIL-V-62038 and the applicable Equipment Preservation Data Sheet (EPDS) for the level(s) specified by the procuring activity (see 6.2).

6. NOTES

6.1 Intended use. Trailers covered by this specification are intended for use in hauling items of construction engineering equipment or similar applications during tactical military operations, and for use under extreme or unusual conditions of climate, weather and terrain.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. First article (preproduction) model, if required (see 3.1).
c. If welding repairs are authorized (see 3.4.5.1).
d. Selection of level of preservation, processing for shipment and storage (see 5.1).

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6.3 Process average. Sampling may be initiated if the process average value for the first twenty trailers inspected is less than the AQL specified in the classification of defects for major and minor defects.

$$\text{Process average} = \frac{\text{Number of defects}}{\text{Number of trailers inspected}} \times 100$$

If the computed process average exceeds the specified AQL, 100 percent inspection shall be performed and continued until such time that the process average for twenty consecutive trailers is less than the specified AQL.

6.4 Deficiency sheet. The Government inspector will verify that a thorough inspection of each trailer is performed by the contractor not only for the listed characteristics, but also for any other departures from good workmanship. The Government inspector will assure that all deficiencies encountered during the inspection are enumerated on the deficiency sheet for the trailer. The defects noted on the deficiency sheet shall contain sufficient description to enable the Government inspector and the contractor's representative to classify the deficiency in accordance with the classification of defects of the trailer specification and definitions contained in MIL-STD-105. Corrective action shall be taken for recurring deficiencies.

6.5 Definitions.

6.5.1 Recurring major deficiencies. A major deficiency (see 6.4), is recurring when the same defect occurs more than once in the same sample, or when the defect occurs in two successive samples. A major defect may be considered recurring when the historical inspection records ("P" chart or approved equivalent) reflect such a condition. Recurring major deficiencies shall be cause for entire lot or lots to be inspected for the recurring deficiencies. The deficiencies shall be corrected when found.

6.5.2 Recurring minor deficiencies. A minor deficiency (see 6.4), is recurring if it occurs more than twice in the same sample, or the defect occurs in four successive samples. Recurring minor deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies and correction shall be accomplished prior to acceptance.

6.6 Recycled major deficiencies. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 3.3).

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6.7 Changes from previous issue. Asterisks are not used in this revision, to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AT

Preparing activity:
Army - AT

Project No. 2330-0613

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE: MIL-T-45333D (AT)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

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